

Invited Review Article

Knowledge mapping of SGLT2-inhibitors and cardiovascular outcomes: A bibliometric analysis



Aysa Rezaakhsh^{a,1,*}, Hadis Iraj^a, Ashot Avagimyan^b, Elena Aghajanova^b, Zinaida Jndoyan^b, Lilia Mirzoyan^c, Waseem Hassan^{d,**}, Solomon Habtemariam^e, Anne Meddahi-Pellé^f, Graciela Pavon-Djavid^g, Abolfazl Barzegri^{h,1}

^a Cardiovascular Research Center, Tabriz University of Medical Sciences, Tabriz, Iran

^b Yerevan State Medical University after M. Heratsi, Yerevan, Armenia

^c Tambov State University after G. R. Derzhavin, Tambov, Russia

^d Institute of Chemical Sciences, University of Peshawar, Pakhtunkhwa, Pakistan

^e Pharmacognosy Research & Herbal Analysis Services, University of Greenwich, Kent, UK

^f Laboratory for Vascular Translational Science, Nanotechnologies for Vascular Medicine and Imaging, University Sorbonne Paris Nord, Paris, France

^g Departement of Hygiene, Security Environment, University Sorbonne Paris Nord, Paris, France

^h Department of Medical Biotechnology, Faculty of Advanced Medical Sciences, Tabriz University of Medical Sciences, Tabriz, Iran

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ABSTRACT

Introduction: Initially developed for the treatment of type 2 diabetes mellitus (T2DM), SGLT2 inhibitors have increasingly been recognized for their substantial cardiovascular, renal, and metabolic benefits. This study undertakes a systematic bibliometric analysis to delineate research trends, quantify advancements, and identify influential studies on SGLT2 inhibitors and cardiovascular outcomes.

Methods: A comprehensive literature search was performed using the Scopus database, resulting in 10,211 documents up to the search date. Both original research articles and reviews (narrative and systematic) were included. Quantitative metrics such as publication count, citation analysis, and authorship were employed. Network visualization tools (e.g., VOSviewer) and bibliometric software (R Studio with bibliometrix packages) were used to analyze collaboration networks, keyword co-occurrence, and thematic clustering.

Results: The analysis revealed a significant upward trend in publication volume, with a notable peak in recent years, indicating a heightened interest in the topic among scientists. The United States and selected European institutions emerged as major contributors, with prominent authors and funding sources influencing the research output. Thematic analysis highlighted a focus on cardiovascular diseases, heart failure, renal outcomes, and mechanistic studies of SGLT2 inhibitor action. Notably, landmark randomized controlled trials have reported significant reductions in cardiovascular mortality, heart failure hospitalizations, and progression of kidney disease.

Conclusion: This study underscores the transformative impact of SGLT2 inhibitors on cardiovascular therapy. The analytical framework highlights the expansion of the field, and the integration of translational research.

* Corresponding author at: Cardiovascular Research Center, Shahid Madani Heart Hospital, Tabriz University of Medical Sciences, Tabriz, Iran.

** Corresponding author at: Institute of Chemical Sciences, University of Peshawar, Peshawar, Pakistan.

E-mail addresses: rezabakhsha@tbzmed.ac.ir (A. Rezaakhsh), Waseem_anw@yahoo.com (W. Hassan).

¹ These authors worked equally as the first authors.

Introduction

The identification of upregulated sodium-glucose co-transporter 2 (SGLT2) proteins in renal tubular epithelial cells has significantly advanced the management of type 2 diabetes mellitus (T2DM).¹ This discovery has led to the development of a class of insulin-independent antihyperglycemic therapeutic agents, collectively called SGLT2 inhibitors. They block the SGLT proteins responsible for sodium and glucose reabsorption in the kidneys, particularly in the proximal convoluted tubules. These agents, including canagliflozin, dapagliflozin, and empagliflozin, preferentially inhibit SGLT2 over SGLT1, resulting in effective glycemic control and additional systemic benefits.^{2,3}

The origins of SGLT2 inhibitors were traced back to the discovery of phlorizin >150 years ago.⁴ Modern SGLT2 inhibitors have been structurally modified to improve their water solubility, bioavailability, and resistance to enzymatic degradation, thereby extending their half-lives and selectivity. SGLT2 proteins are responsible for over 97 % of glucose reabsorption in the early proximal tubule, and their inhibition leads to improved glycemic control at all stages of T2DM. Importantly, the efficacy of SGL2 inhibitors diminishes when glucose levels fall below a certain threshold, thereby preventing hypoglycemia.^{5,6}

In addition to glycemic control, SGLT2 inhibitors confer renoprotective effects by increasing fluid and electrolyte delivery to the macula densa, activating tubuloglomerular feedback, and reducing hyperfiltration, albuminuria, and hypoxia in renal tissues. Their insulin-independent mechanism of action allows for sustained efficacy even as β -cell function declines or insulin resistance progresses. Physiologically, SGLT2 inhibitors initially reduce plasma volume and estimated glomerular filtration rate (eGFR), followed by a modest increase in eGFR, resulting in hemodynamic changes that prevent renal dysfunction.⁷ These agents also promote glucosuria, leading to caloric loss and significant weight reduction, likely owing to enhanced lipolysis and fatty acid oxidation.^{8,9}

SGLT2 inhibitors exhibit notable cardiometabolic effects. By inhibiting glucose and sodium reabsorption, they induce natriuresis and osmotic diuresis, thereby lowering plasma volume and blood pressure.⁷ Mechanistically, these effects are attributed to improved ventricular loading conditions, enhanced cardiac metabolism, promotion of ketone-body utilization, inhibition of myocardial Na^+/H^+ exchange, and modulation of adipokine and proinflammatory cytokine production. SGLT2 inhibitors may also stimulate erythropoiesis and improve oxygen-carrying capacity, which is advantageous for patients.^{10,11}

In the context of cardiovascular disease (CVD), SGLT2 inhibitors have demonstrated efficacy in reducing cardiovascular risk in patients with heart failure (HF), atherosclerosis, and pulmonary hypertension.^{12,13} These benefits are thought to arise from modulation of mitochondrial function, cytoskeletal proteins, and fibrosis markers. However, the effectiveness of different SGLT2 inhibitor-based prevention strategies against major adverse cardiovascular events remains unclear.¹⁴

HF affects over 64 million individuals globally and is associated with elevated morbidity and mortality rates. SGLT2 inhibitors have shown considerable promise in improving survival, quality of life, cardiac function, and exercise capacity in patients with HF regardless of their glycemic status. Major cardiovascular outcome trials (CVOTs), such as EMPA-REG OUTCOME, CANVAS, and DECLARE-TIMI 58, have consistently demonstrated reductions in major adverse cardiovascular events and hospitalizations in patients with HF.¹⁴ These benefits are primarily attributed to hemodynamic effects, including natriuresis, enhanced diastolic function, and renoprotective effects. SGLT2 inhibitors, particularly dapagliflozin and empagliflozin, are recommended as first-line medications for patients with HF across the entire spectrum of left ventricular ejection fraction, irrespective of the presence of T2DM (32). A similar trend was observed in chronic kidney disease.¹⁵

Current research has focused on evaluating the efficacy of treatments across diverse patient populations, including those experiencing cardiac dysfunction related to cancer therapy and various HF phenotypes.¹⁶ Additionally, there is a significant interest in exploring the potential influence of genetic polymorphisms on therapeutic responses. Emerging evidence also indicates that early administration of SGLT2 inhibitors following myocardial infarction is associated with reduced infarct size, decreased post-myocardial infarction fibrosis area, and improved cardiac outcomes.¹⁷ SGLT2 inhibitors represent a significant advancement in the management of T2DM, HF, and CVD, and ongoing trials are expected to further elucidate their multifaceted benefits.¹⁸ In this context, the multimodal pleiotropic potential of SGLT2 inhibitors should be highlighted, particularly their neuroprotective, hepatoprotective, and oncoprotective effects.¹⁹

This study aimed to assess both current and prospective research trends in the field by analyzing the publication details of SGLT2 inhibitors that affect HF using bibliometric analysis tools.

Methods

This study used the Scopus database, which is one of the largest and most comprehensive abstract and citation databases in the peer-reviewed literature. An open timeframe was applied (excluding 2025), meaning that no restrictions were placed on the year of publication, allowing for the inclusion of all relevant documents indexed in Scopus up to the search date. To ensure the quality and relevance of the data, only the documents classified as articles or reviews were included in the analysis.

The following search string was used: “Sodium-Glucose Cotransporter 2 Inhibitors” OR “SGLT2i” OR “SGLT2-Inhibitors” OR “Gliflozins” OR “Flozins” OR “Empagliflozin” OR “Dapagliflozin” OR “Sotagliflozin” OR “Canagliflozin” OR “Ertugliflozin” OR “Ipragliflozin” OR “Bexagliflozin” OR “Tofogliflozin” OR “Luseogliflozin” AND “Cardiovascular disease” OR “CVD” OR “Heart failure” OR “HF” OR “Myocardial Infarction” OR “MI” OR “Percutaneous Coronary Intervention” OR “PCI” OR “Cardiovascular Risk Reduction.” Two complementary approaches were employed in the data collection and analysis process.

1. First, a search string was applied to the title, abstract, and keyword (TITLE-ABS-KEY) fields in Scopus to identify the full set of relevant publications that matched the criteria.
2. Second, the top 100 most-cited documents were separately identified and downloaded from the resulting dataset to provide citation-based insights into the most influential research within the field.

All data were exported in the CSV format from Scopus. The datasets were subsequently organized, cleaned, and analyzed using Microsoft Excel for preliminary inspection, VOSviewer for network visualization (e.g., co-authorship and keyword co-occurrence), and R Studio (using the *bibliometrix* package) for advanced bibliometric and thematic analysis.

Results and discussion

A total of 10,211 documents were retrieved from the Scopus database using a defined search strategy. Among these, 6,388 were original research articles and 3,823 were review papers, indicating substantial scholarly attention from both primary investigators and review authors.

The temporal distribution of publications has increased markedly in recent years. The number of publications is expected to peak in 2024, with 2,479 documents, 1,928 in 2023, 1,765 in 2022, and 1,403 in 2021. This reflects a strong upward trend in the research activities related to SGLT2 inhibitors and cardiovascular outcomes. Yearly publication details are presented in Fig. 1.

The data also show a steady increase from 2014 onward, with minor research activity in the earlier years. For instance, only one document each was published in 2007 and 2009, followed by a small, but gradually increasing output until 2014. Since then, the number of publications has accelerated, surpassing 1,000 annual publications in 2021 and doubling between 2020 (886) and 2024 (2,479). This pattern highlights the growing scientific and clinical interest in the cardiometabolic implications of SGLT2 inhibitors over the past decade.

An analysis of the most prolific contributors to the field revealed that Javed Butler led authorship, contributing to 244 publications. He was closely followed by Silvio E. Inzucchi and John J.V. McMurray, each with 183 publications, whereas Scott D. Solomon and Subodh Verma contributed to 153 and 150 publications, respectively. These authors are central figures in the cardiovascular and diabetes research landscape, particularly in the studies on SGLT2 inhibitors.

In terms of institutional affiliations, Harvard Medical School emerged as the most active, contributing 522 publications, followed by Brigham and Women's Hospital (504 publications), and the University of Toronto (422). Other major contributors included the University of Glasgow (330 publications) and St. Michael's Hospital in Toronto (271 publications). These institutions appear to serve as leading hubs for research on cardiometabolic therapy and its outcomes.

At the country level, the United States dominated research output with 3,463 publications, significantly ahead of the United Kingdom (1,420 publications) and China (1,128 publications). Italy and Germany also demonstrated strong activities with 982 and 969 publications, respectively. These findings suggest that research in this area is largely driven by institutions and collaboration in high-income countries.

An analysis of funding sources showed that AstraZeneca was the most prominent sponsor, supporting 766 publications and likely reflecting the company's vested interest in the development and evaluation of SGLT2 inhibitors. Other major sponsors included the National Institutes of Health (NIH) with 715 publications, Boehringer Ingelheim (683), U.S. Department of Health and Human Services (593), and Novo Nordisk (436), highlighting a strong mix of both industry and public sector investments in this area.

Regarding publication venues, Diabetes Obesity and Metabolism led to 306 publications, followed by Cardiovascular Diabetology (281), the Journal of Clinical Medicine (194), the European Journal of Heart Failure (183), and Circulation (174). These journals



Fig. 1. The list of the per-year number of publications.

represent the key platforms through which pivotal findings in this domain are disseminated. Data for the top contributors are presented in **Supplement 1**.

Research focus of all papers

The top 160 terms were extracted from the Scopus database based on the number of appearances. To enhance thematic clarity, simple and generic terms such as “male,” “female,” “article,” and other broadly descriptive labels were excluded. The remaining, more specific terms were systematically organized into seven thematic categories, which reflect the key focus areas of the literature. These categories are presented below, along with descriptions of the main themes emerging from each category.

Diseases and conditions

A central theme in the literature revolves around major chronic and metabolic diseases, with a strong emphasis on cardiovascular and renal conditions. HF ($n = 6181$), non-insulin dependent diabetes mellitus ($n = 5586$), and cardiovascular disease ($n = 4294$) appeared most frequently, indicating the dominance of cardiometabolic comorbidities. Other recurrent conditions included type 2 diabetes ($n = 3712$), hypertension ($n = 2267$), chronic kidney failure ($n = 1954$), and complications, such as cardiovascular mortality ($n = 1648$), hypoglycemia ($n = 1285$), obesity ($n = 1257$), and atrial fibrillation ($n = 1103$). This category highlights the interconnected burden of diabetes, heart disease, and kidney dysfunction on clinical research and patient care.

Medications and therapies

This category presents a significant focus on pharmacological interventions, particularly those for diabetes and HF. The high frequency of terms related to SGLT2 inhibitors—such as empagliflozin ($n = 3605$), dapagliflozin ($n = 3259$), and canagliflozin ($n = 2063$) illustrates the growing importance of this drug class. Other common therapies included metformin ($n = 2552$), insulin ($n = 1968$), GLP-1 receptor agonists such as liraglutide ($n = 1116$), and DPP-4 inhibitors such as sitagliptin ($n = 739$). Additionally, drugs affecting the renin-angiotensin system and beta blockers also appeared frequently, underscoring their role in cardiovascular protection. This category reflects the emphasis on multidrug approaches for glycemic control and cardiovascular outcomes.

Outcomes and clinical measures

The literature strongly focuses on clinical endpoints and therapeutic monitoring. Frequently mentioned terms included cardiovascular risk ($n = 2018$), hemoglobin A1C ($n = 1992$), estimated glomerular filtration rate ($n = 1751$), and drug efficacy ($n = 1642$). Measures such as mortality ($n = 1362$), risk factors ($n = 1017$), all-cause mortality ($n = 1110$), and clinical outcome ($n = 1209$) reflect the evaluation of intervention effectiveness and patient prognosis. The repeated appearance of glycemic control ($n = 1444$), body weight ($n = 593$), and insulin resistance ($n = 662$) indicated a strong interest in metabolic improvements and their impact on survival and quality of life.

Study designs and methodology

This group highlights the methodological rigor of the analyzed research. The frequent use of randomized controlled trials ($n = 1426$, topic; $n = 1034$, format), cohort analyses ($n = 1275$), and systematic reviews ($n = 744$) indicated a preference for evidence-based evaluation. The presence of terms such as observational study ($n = 649$), meta-analysis ($n = 682$), and multicenter study ($n = 613$) revealed methodological diversity and collaborative approaches. The appearance of practice guidelines ($n = 1085$) and priority journals ($n = 1116$) also suggests that many findings are integrated into clinical recommendations or are published in influential sources.

Laboratory and physiological measures

This category includes key biomarkers and physiological indicators used for diagnosis and monitoring. Frequent mentions of glucose ($n = 1895$), blood pressure ($n = 838$), and creatinine ($n = 957$) levels demonstrated the importance of tracking metabolic and renal parameters. Other markers such as albuminuria ($n = 664$), LDL cholesterol ($n = 712$), and natriuretic peptides (BNP, $n = 602$) are commonly used in cardiovascular and renal outcome studies. These terms suggest a systematic reliance on quantitative biomarkers for risk stratification and treatment optimization.

Biological and molecular concepts

This thematic area delves into the mechanistic understanding of disease and drug action. Frequently cited concepts included pathophysiology ($n = 1416$), metabolism ($n = 1338$), oxidative stress ($n = 731$), and inflammation ($n = 688$). The presence of biological markers ($n = 614$) and drug-target pathways, such as the renin-angiotensin aldosterone system ($n = 599$), reveals an ongoing interest in uncovering the molecular underpinnings of treatment responses and disease progression.

Others (Descriptive, Demographic, Technical)

This category included terms with broader contextual and descriptive dimensions. High frequencies of hospitalization ($n = 2522$), complication ($n = 2491$), and prevalence ($n = 1022$) reflect the real-world burden and epidemiology of chronic diseases. Comorbidity ($n = 975$), body mass ($n = 1211$), and quality of life ($n = 826$) indicated the importance of patient-centered outcomes. The appearance of unclassified drugs, benzoyl derivatives, and sodium glucose co-transporter 2 illustrates their technical and chemical specificities. Finally, inclusion of lifestyle modification ($n = 579$) showed interest in non-pharmacological approaches, albeit to a lesser extent.

The main limitation of this approach is that we focused exclusively on the 160 most frequently occurring terms extracted from the Scopus database. Although this selection provided valuable insights into the most commonly studied concepts, it excluded other potentially relevant terms that could have contributed to a more comprehensive understanding of the research landscape. Moreover, it is important to note that it is not feasible to incorporate every possible term from the entire corpus of research articles, because doing so would require significantly larger and more complex analyses. A broader inclusion of terms would have risked introducing noise into the data, reducing the clarity of the key trends and themes. Thus, although the exclusion of additional terms may limit the breadth of this study, it is necessary to balance the feasibility of delivering meaningful insights.

The top 100 most cited documents

An analysis of the top 100 most-cited documents from 2014 to 2024 revealed substantial citation impacts and notable temporal trends. These documents were published across 32 different sources, and included 65 original articles and 35 reviews. The average age

Table 1

Based on the number of publications, the top authors, universities, countries, sponsors and journals in all publications.

S#	Author	Number of Publications
1	Butler, J.	244
2	Inzucchi, S.E.	183
3	McMurray, J.J.V.	183
4	Solomon, S.D.	153
5	Verma, S.	150
6	Kosiborod, M.N.	149
7	Vaduganathan, M.	135
8	Langkilde, A.M.	134
9	Packer, M.	131
10	Lam, C.S.P.	119
S#	University	Number of Publications
1	Harvard Medical School	522
2	Brigham and Women's Hospital	504
3	University of Toronto	422
4	University of Glasgow	330
5	St. Michael's Hospital, Toronto	271
6	Yale School of Medicine	271
7	Universitair Medisch Centrum Groningen	267
8	Rijksuniversiteit Groningen	265
9	Imperial College London	253
10	George Institute for Global Health	244
S#	Country	Number of Publications
1	United States	3463
2	United Kingdom	1420
3	China	1128
4	Italy	982
5	Germany	969
6	Canada	845
7	Japan	836
8	Australia	586
9	Sweden	548
10	Netherlands	545
S#	Sponsors	Number of Publications
1	AstraZeneca	766
2	National Institutes of Health	715
3	Boehringer Ingelheim	683
4	U.S. Department of Health and Human Services	593
5	Novo Nordisk	436
6	National Natural Science Foundation of China	364
7	Novartis	315
8	Merck	295
9	Ministry of Science and Technology of the People's Republic of China	286
10	Eli Lilly and Company	285
S#	Journal	Number of Publications
1	Diabetes Obesity And Metabolism	306
2	Cardiovascular Diabetology	281
3	Journal Of Clinical Medicine	194
4	European Journal Of Heart Failure	183
5	Circulation	174
6	Esc Heart Failure	166
7	Frontiers In Cardiovascular Medicine	159
8	European Heart Journal	144
9	Diabetes Care	140
10	Diabetes Therapy	139

of the documents was 5.3 years, reflecting a mix of both recent and older influential works. Each document received an average of 1,253 citations, underscoring the significant academic attention these publications garnered. Table 1 presents the descriptive data for the 100 most-cited papers are presented in Table 1.

The earliest year, 2014, had only two papers, with a modest mean total citation (TC) per article of 820.00 and a mean citation per year (TCpY) of 68.33. 2015 emerged as a standout year, with four documents averaging 3,175.00 citations each and a TCpY of 288.64, the second highest in the dataset. From 2016 to 2021, both the volume and impact of papers increased, peaking in 2021 with 18 publications, averaging 1,391.83 citations per paper, and a TCpY of 278.37. The year 2019 was also prominent, with 16 papers and the highest volume-adjusted mean TCpY of 209.86, indicating a robust influence, despite being more recent. In recent years, such as 2022 and 2023, high citation momentum has been maintained, with TCpY values of 249.60 and 294.76, respectively, suggesting strong early uptake. The sole 2024 paper stood out with a remarkable TCpY of 468.00, although its impact remains to be confirmed over time, owing to its recency Table 2.

In summary, although the annual publication of top-cited papers has decreased over time, the citation density per article has generally increased, particularly in recent years, indicating a trend towards fewer but more impactful publications in the field.

The top contributors: In the top 100 most cited documents

The top 100 most-cited documents were the result of contributions from 3,521 authors, reflecting a highly collaborative research landscape. The average number of co-authors per document was exceptionally high at 45.3, indicating extensive multi-author and often multi-center collaborations. Only one paper was authored individually, emphasizing the rarity of single-author contributions to this body of literature. Additionally, 60 % of the documents involved international co-authorship, underscoring the global integration of research efforts in this field.

Several authors have contributed to their scientific contributions. McMurray JJV emerged as the most impactful contributor, with an h-index and g-index of 12 and a cumulative 23,976 citations from 12 publications since 2018, yielding an m-index of 1.5. McGuire closely followed 11 publications, 13,709 citations, and a higher m-index of 1.571, indicating a slightly more rapid accumulation of impactful works since 2019. Inzucchi SE, active since 2015, contributed 10 documents amassing 24,312 citations and maintained a strong scholarly presence, with an h-index of 10 and an m-index of 0.909.

Institutional affiliation also played a significant role. Harvard Medical School led with 36 articles, followed by the University of Toronto with 26 articles, reflecting a strong institutional involvement in highly cited work. Geographically, the United States dominated 56 publications, followed by the United Kingdom (37) and Canada (24), indicating a concentration of influential research within North America and selected parts of Europe.

In terms of publication sources, the New England Journal of Medicine hosted the largest number of top-cited documents (15 articles), followed by Diabetes Care (12), Circulation (11), and The Lancet (10). These findings suggest that a significant portion of high-impact research in this field has been published in top-tier high-visibility journals. The list of the top 100 authors (along with the number of publications, total citations, h-index, g-index, and m-index), top 100 affiliations, all countries, and all sources is presented in **Supplementary Tables 1-5**, respectively. The collaboration networks for the authors, departments, and countries are presented in **Supplementary Figs. 1-3**, respectively.

The research focus of the top 100 most cited papers

To understand the core research focus of the top 100 most-cited papers, we also employed the bi-gram and tri-gram dynamics approaches. This technique analyzes frequently occurring two-word (bi-gram) and three-word (tri-gram) phrases across publication years, allowing us to uncover the evolving thematic patterns within the literature.

The bi-gram approach enhances contextual understanding by capturing specific conceptual pairings, rather than isolated keywords. It reveals how critical terms—such as “heart failure” or “clinical practice”—consistently co-occur, offering more meaningful insights into dominant research topics. By tracing the frequency of these pairs over time, we can clearly track the emergence, persistence, and decline of key themes and therapeutic interests.

The tri-gram approach builds on this foundation by identifying more complex and nuanced expressions, such as “reduced ejection fraction”, “chronic kidney disease”, or “sodium-glucose cotransporter inhibitors”. These longer phrases offer greater specificity,

Table 2
The per year publications and citation details of the top 100 most cited documents.

Year	Mean Total Citations per Article	Number of Publications	Mean Total Citations per Year	CitableYears
2014	820.00	2	68.33	12
2015	3,175.00	4	288.64	11
2016	1,084.17	6	108.42	10
2017	1,445.40	5	160.60	9
2018	773.00	9	96.62	8
2019	1,469.00	16	209.86	7
2020	1,163.38	16	193.90	6
2021	1,391.83	18	278.37	5
2022	998.38	16	249.60	4
2023	884.29	7	294.76	3
2024	936.00	1	468.00	2

particularly in specialized clinical domains, where precise terminology reflects distinct diagnostic categories, treatment standards, or pharmacological targets. Trigram analysis is especially powerful for differentiating overlapping research areas and pinpointing the influence of specific guidelines or consensus statements. The details of the bigram and trigram dynamics are shown in Figs. 2 and 3, respectively.

Cardiovascular and HF focus

A consistent increase in terminology related to *heart failure* (e.g., "heart failure" = 36 in 2024) and *ejection fraction* suggests a dominant focus on *cardiac function*, particularly heart failure with a reduced or preserved ejection fraction. This is supported by the frequent appearance of terms such as chronic heart failure, reduced ejection fraction, preserved ejection fraction, heart failure association, and joint guideline terms involving cardiology and heart associations. The integration of these terms in both bi- and tri-grams indicates a steady emphasis on *clinical guidance, classification, and management of cardiac dysfunction*.

Diabetes and glycemic control

Terms such as "type diabetes," "diabetes mellitus," "type diabetes mellitus, and "glycemic treatment standards" glycemic treatment standards have shown continuous growth, with *type 2 diabetes management* emerging as a central theme. Organizations, such as the *American Diabetes Association (ADA)* and their "position statements," "consensus reports," and collaborations with the *European Association for the Study of Diabetes (EASD)* reflect the growing standardization of care. These themes also overlapped in the bi- and tri-gram listings, reinforcing the centrality of *guideline-based diabetes management*.

Kidney and renal outcomes

"Kidney disease," "chronic kidney," "chronic kidney disease," and "renal outcomes" have gained attention, especially after 2018, emphasizing the shift toward *renal risk assessment and protection*, particularly in diabetic patients. The increase of "KDIGO clinical practice" references also suggests growing adherence to nephrology-specific consensus frameworks.

SGLT inhibitors and glucose Co-transporters

There is clear, sustained attention to "SGLT inhibitors," "sodium-glucose cotransporter inhibitors," and variations such as "SGLT-inhibitors" or "cotransporter-inhibitors." These terms collectively underscore the emergence of *SGLT2 inhibitors* as cornerstones in the management of *cardiovascular and renal complications in diabetes*. The frequent mention of these terms across both the bi-gram and tri-gram sets highlights their crosscutting relevance.

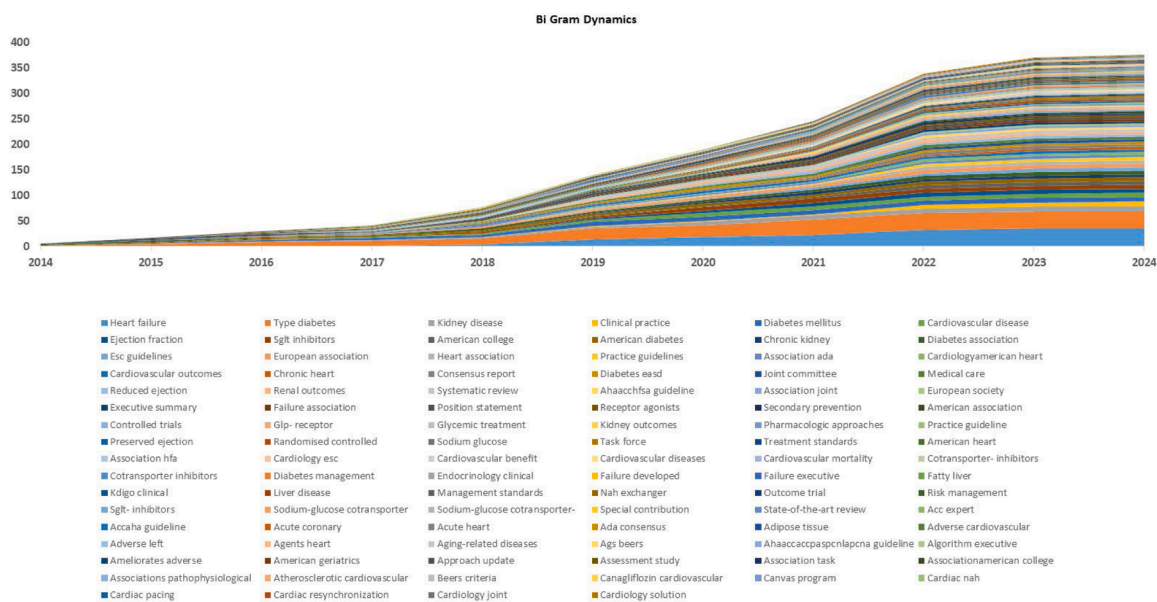


Fig. 2. Bi-gram of the top 100 most cited papers.

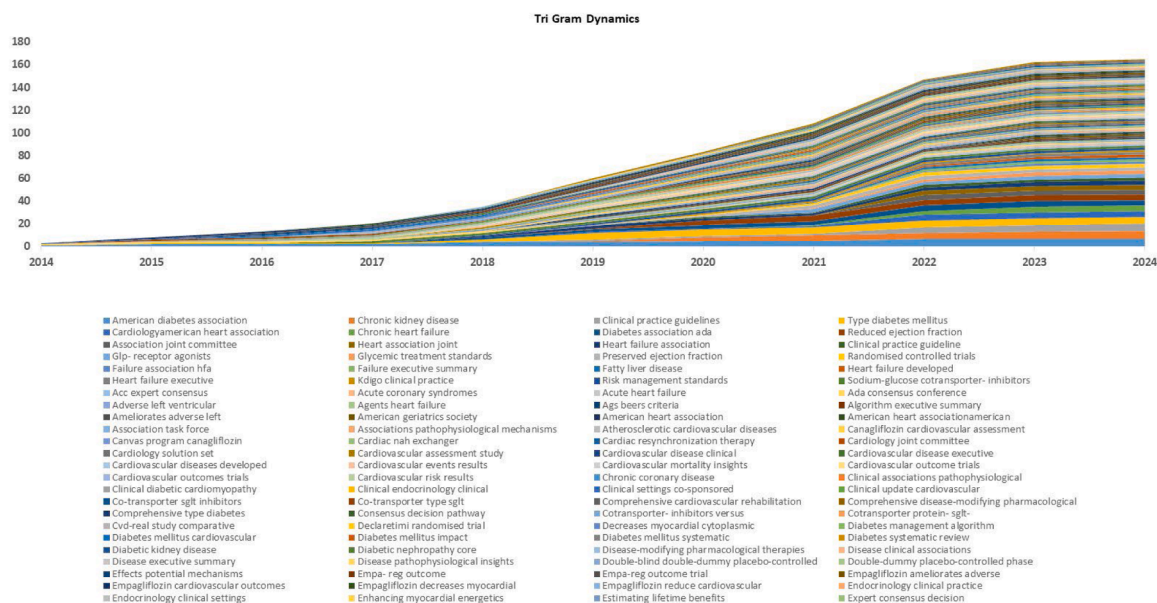


Fig. 3. Tri gram of the top 100 most cited papers.

Professional societies and clinical guidelines

Institutional references - e.g., “American Diabetes Association,” “American Heart Association,” “European Society,” and terms like “clinical practice guidelines,” “ESC guidelines,” “consensus report,” and “executive summary”—indicate a strong alignment of research with *evidence-based recommendations*. These terms often appear in both bi- and tri-grams, showing their embedded role in framing disease-management strategies.

Cardiovascular outcomes and risk management

Terms such as “cardiovascular disease,” “cardiovascular mortality,” “cardiovascular benefit,” “risk management,” and “secondary prevention” secondary prevention have received increasing attention in *long-term outcome trials* and *preventive strategies*. This is also associated with the use of drugs such as GLP-1 receptor agonists, indicating a therapeutic shift toward *cardiometabolic protection*.

Trials, reviews, and standardization

Frequent mentions of “randomized controlled trials,” “systematic review,” and “state-of-the-art review” reflect a maturing research landscape focused on *synthesizing evidence*. Meanwhile, terms like “practice guideline,” “treatment standards,” and “medical care” show increasing implementation of *standardized treatment frameworks*.

Emerging and niche topics

There’s growing interest in “fatty liver disease,” “adipose tissue,” “cardiac resynchronization,” and “cardiac pacing,” suggesting expansions into *comorbid conditions* and *specialized cardiac therapies*. Although less dominant, their steady appearance signals have evolved into subspecialty interests.

The research focus of the top 10 most cited publications

The title co-word analysis of the top ten most-cited publications revealed a strong focus on SGLT2 inhibitors and cardiovascular-renal outcomes in diabetes and HF management.

Zinman et al. (9443 citations) investigated the effects of empagliflozin on cardiovascular outcomes and mortality in type 2 diabetes patients, directly emphasizing the relationship between a specific SGLT2 inhibitor and mortality reduction.²⁰ McDonagh et al. (8198 citations) provided the 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic HF, highlighting updated clinical recommendations based on emerging therapies, including SGLT2 inhibitors.²¹ McMurray et al. (4951 citations) evaluated the effects of dapagliflozin in patients with HF and reduced ejection fraction, focusing on the clinical benefits of HF beyond glucose control.²²

Wiviott et al. (4672 citations) focused on dapagliflozin and cardiovascular outcomes in patients with type 2 diabetes, reinforcing the emerging cardioprotective role of SGLT2 inhibitors.²³ Perkovic et al. (4406 citations) explored canagliflozin and its effects on renal

outcomes in type 2 diabetes with nephropathy, drawing attention to the prevention of kidney disease progression.²⁴ Packer et al. (3501 citations) studied the cardiovascular and renal outcomes of empagliflozin in patients with HF, reflecting the continuation of evaluating SGLT2 inhibitors in HF populations.²⁵

Heerspink et al. (3346 citations) reported the use of dapagliflozin in patients with CKD, indicating a focus on renal protection irrespective of diabetes status.²⁶ Neal et al. (3252 citations) examined canagliflozin and cardiovascular and renal events in type 2 diabetes, strengthening the evidence for dual-organ protection.²⁷ Cosentino et al. (3206 citations) presented the 2019 ESC Guidelines on diabetes, pre-diabetes, and cardiovascular diseases developed with EASD collaboration, integrating SGLT2 inhibitor outcomes into standard care recommendations.²⁸ Finally, Anker et al. (2948 citations) analyzed the benefits of empagliflozin in HF with a preserved ejection fraction, expanding the therapeutic scope of SGLT2 inhibitors to a broader HF population.²⁹

Across these highly cited studies, the consistent co-word pattern between drug names (empagliflozin, dapagliflozin, and canagliflozin), disease states (type 2 diabetes, HF, and CKD), and outcomes (cardiovascular, renal, and mortality) indicated a central scientific focus on redefining treatment paradigms through SGLT2 inhibition.

Future perspectives

This study provides a comprehensive overview of the major trends, key contributors, and thematic priorities that have shaped the most influential research on cardiometabolic health in the past decade. By focusing on the top 100 most-cited publications and the most frequently occurring terms, we identified a strong concentration of research on SGLT2 inhibitors, cardiovascular and renal outcomes, and the integration of evidence into clinical practice guidelines. The findings highlight not only the scientific and therapeutic evolution within diabetes and HF management but also the critical role of international collaboration, multidisciplinary research teams, and high-visibility journal outlets in achieving a significant citation impact. While the analysis intentionally focused on the most prominent terms and documents for clarity and feasibility, it successfully captured dominant scientific directions and emerging areas of interest. These insights offer valuable guidance for future research strategies, collaboration models, and clinical applications aimed at advancing the care for cardiometabolic diseases.

This study has several important implications for the researchers, clinicians, and policymakers involved in cardiometabolic health research. First, by systematically analyzing the top 100 most-cited papers and the most frequently occurring terms, this study provides a clear and evidence-based understanding of the major themes that have shaped the field over the past decade. This highlights the central role of SGLT2 inhibitors in redefining treatment strategies for type 2 diabetes, HF, and CKD, emphasizing a major therapeutic shift toward multiorgan protection beyond glycemic control.

Second, the analysis underscores the growing influence of clinical practice guidelines and consensus statements from leading professional societies, suggesting that successful integration into clinical recommendations is a key driver in high-impact research. This finding reinforces the need for future research initiatives to closely align with evolving guideline priorities to enhance both clinical relevance and citation impact.

Third, strong international collaboration networks revealed by authorship and institutional analyses highlight the increasing importance of global research partnerships in advancing high-impact science. Institutions and researchers aiming to amplify their academic influence should prioritize multinational and multicenter collaborations.

Fourth, the high average number of co-authors per article and the dominance of publications in top-tier journals suggest that producing widely cited work often requires coordinated, large-scale efforts and the strategic targeting of high-visibility outlets.

Finally, a thematic analysis of emerging topics, such as fatty liver disease and specialized cardiac therapies, suggests fertile areas for future investigation. Researchers can use these insights to identify promising subfields that are gaining momentum but may still offer opportunities for pioneering contributions.

Conclusion

In summary, this study provides a strategic map of research priorities, collaborative patterns, and publication practices that have defined the most influential work in the field, offering valuable guidance for shaping future research directions, collaborations, and dissemination strategies.

Limitations

This study has several limitations. First, the analysis was based solely on the data extracted from the Scopus database. While Scopus is a comprehensive and widely recognized source for bibliometric studies, its exclusive reliance on a single database may have led to the omission of relevant publications indexed elsewhere. Although we carefully designed the search string to capture the core research output, any addition, deletion, or modification of keywords or Boolean operators could alter the results and influence the selection of articles and overall findings.

Second, this study was purely quantitative. It focuses on citation metrics, keyword occurrences, and authorship patterns without assessing the qualitative aspects of the research, such as clinical relevance, methodological rigor, or societal impact. Future systematic reviews and expert-driven assessments are encouraged to complement these findings by exploring the clinical implications and real-world contributions of the most cited studies.

By acknowledging these limitations, we aimed to provide a transparent account of the study's methodological scope and encourage further research that could refine and expand our results.

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Authors' contributions

AR conceived and designed the study; WH and HI prepared the first draft; GPD, AMP, AA, and SH revised the manuscript; WH and AB performed the bibliometric analysis. All authors have read and approved the final version of the manuscript.

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The datasets supporting the conclusions of this article are included within the article.

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Supplementary materials

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